

HORNSBY GIRLS HIGH SCHOOL



Mathematics Advanced

Year 11 Assessment 2 Term 2 2023

STUDENT NUMBER: _____

General Instructions

- Reading Time – 5 minutes
- Working Time – 60 minutes
- Write using black pen
- Calculators approved by NESA may be used
- In Questions 6 – 11, show relevant mathematical reasoning and/or calculations
- Marks may be deducted for untidy and poorly arranged work
- Do not use correction fluid or tape
- Do not remove this paper from the examination
- Students are permitted to refer to one double-sided A4 page of handwritten topic summaries which they prepared prior this assessment

Total marks – 53

Section I Pages 2 – 3

5 marks

Attempt Questions 1 – 5

Allow about 8 minutes for this section

Answer on the Objective Response Answer Sheet provided

Section II Pages 4 – 15

48 marks

Attempt Questions 6 -11

Allow about 52 minutes for this section

Write your answers in the space provided.

Question	1-5	6	7	8	9	10	11	Total
Total	/5	/8	/9	/9	/8	/7	/7	/53

This assessment task constitutes 35% of the Preliminary Higher School Certificate Course School Assessment

Section I

5 marks

Attempt Questions 1 – 5

Allow about 8 minutes for this section

Use the Objective Response answer sheet for Questions 1 – 5

1 Which of the following is **NOT** equal to $\sin 30^\circ$?

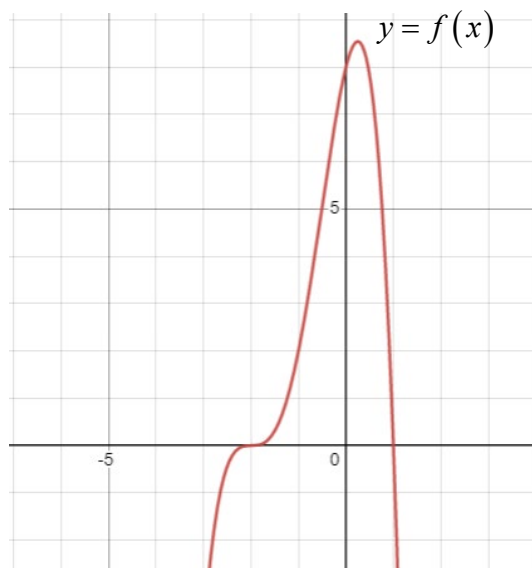
A. $\cos 60^\circ$

B. $\sin(-30^\circ)$

C. $-\sin 210^\circ$

D. $\sin \frac{5\pi}{6}$

2 The graph of $y = f(x)$ is shown on the number plane below.



Which of the following could be the equation of this graph?

A. $y = (x - 1)(2 - x)^3$

B. $y = (x + 1)(x + 2)^3$

C. $y = (1 - x)(x + 2)^3$

D. $y = (1 + x)(x - 2)^3$

- 3 What are the coordinates of the vertex of the parabola $y = -(x - 3)^2 + 2$?
- A. $(3, 2)$
 - B. $(3, -2)$
 - C. $(-3, 2)$
 - D. $(-3, -2)$
- 4 The graph of a particular quadratic equation $y = ax^2 + bx + c$ is a concave down parabola with no x -intercepts. Which of the following gives the correct values for a and the discriminant, Δ ?
- A. $a > 0, \Delta > 0$
 - B. $a > 0, \Delta < 0$
 - C. $a < 0, \Delta > 0$
 - D. $a < 0, \Delta < 0$
- 5 Let $h(x) = f(g(x))$ where the function $f(x)$ is an even function and the function $g(x)$ is an odd function.
- Which of the following statements is **true**?
- A. $h(x)$ is an even function.
 - B. $h(x)$ is an odd function.
 - C. $h(x)$ is both an even and an odd function.
 - D. $h(x)$ is neither an even nor an odd function.

End of Section I

Section II

50 marks

Attempt Questions 6-11

Allow about 52 minutes for this section

Question 6 (8 marks)

- (a) Find the exact value of $\sin 120^\circ$. 1

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- (b) Find the exact value of $\cos \frac{5\pi}{4}$. 1

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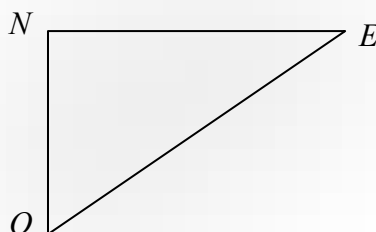
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- (c) A ship sails from a point O due north for 8 km to a point N , and then sails due east for 12 km to a point E . What is the bearing of O from E , to the nearest degree?

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NOT TO
SCALE

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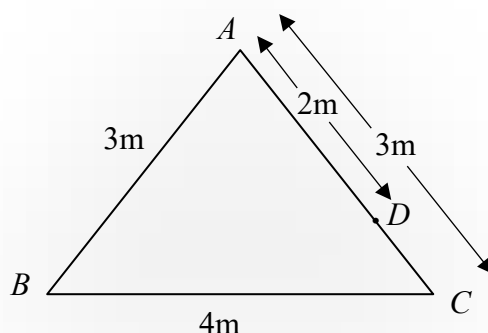
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Question 6 continues on page 5

Question 7 (9 marks)

- (a) In the diagram $AB=3\text{m}$, $AC=3\text{m}$ and $BC=4\text{m}$.
The point D on AC is such that $AD=2\text{m}$.



NOT TO
SCALE

- (i) Find the value of $\angle A$, to the nearest minute.

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- (ii) Calculate the length of BD , correct to 2 decimal places.

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- (iii) Find the area of $\triangle ABD$, correct to 2 decimal places.

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Question 7 continues on page 7

- (b) A plane is travelling at a constant speed of 600 km/h. At a certain instant it is directly overhead. Twelve seconds later, its angle of elevation is 68° . Find the height of the plane above the ground to the nearest metre.

[illegible]

End of Question 7

Question 8 (9 marks)

- (a) Given that $f(x) = x^2$ and $g(x) = \sqrt{x}$, state the domain of $f(g(x))$. **1**

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- (b) Find the equation of the cubic function $f(x) = kx^3 + c$ if it passes through the point $(-1, 2)$ and has y intercept -3 . **2**

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- (c) Find the points of intersection of the graphs of $y = x^2 - x$ and $y = 2x - 2$. **3**

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Question 8 continues on page 9

(d) Find the points of intersection of the graphs of $4x - 3y = 22$ and $3x + 4y = 29$.

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End of Question 8

Question 9 (8 marks)

(a) Solve $\left| \frac{x}{3} - 2 \right| - 1 = 0$.

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(b) $A(-1, 3)$ is a vertex of the rectangle $ABCD$ whose side AD has equation $x + 2y - 5 = 0$.
Find the equation of the side AB .

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Question 9 continues on Page 11

Question 9 (continued)

- (c) (i) If $f(x) = \frac{1}{x-2} + 1$, find $f(3)$.

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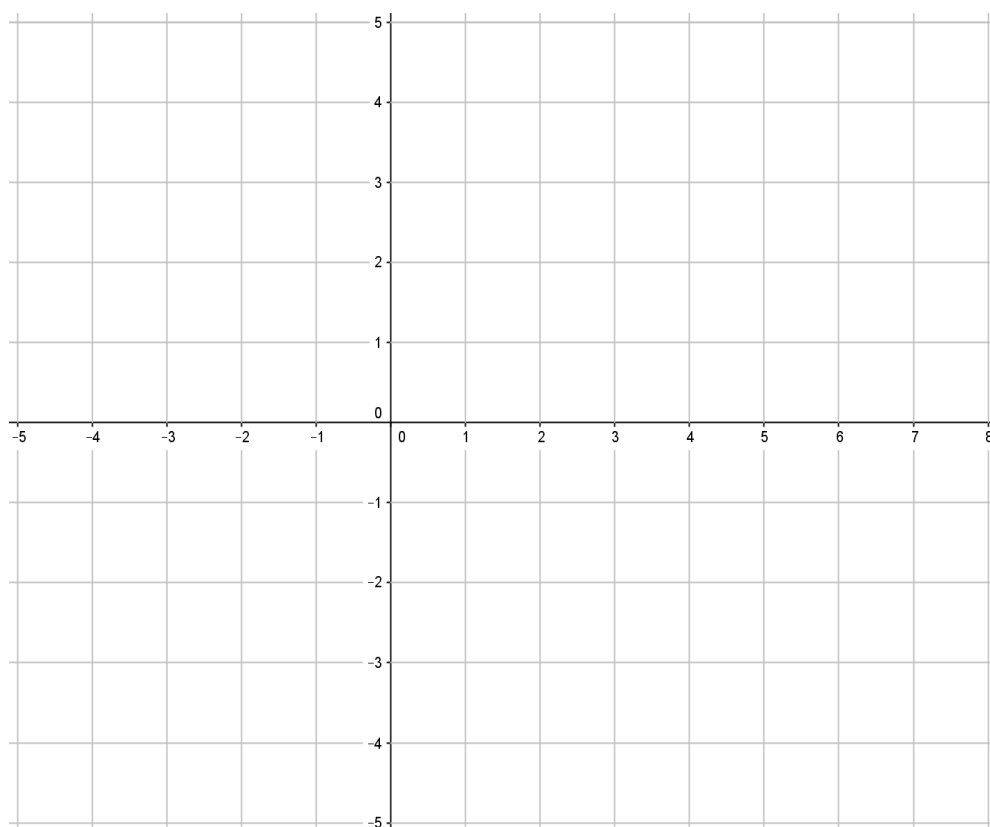
- (ii) State the domain of the function $f(x) = \frac{1}{x-2} + 1$.

1

.....

- (iii) Sketch the graph of the function $f(x) = \frac{1}{x-2} + 1$, showing the x and y intercepts and the equations of any asymptotes.

2



End of Question 9

Question 10 (7 marks)

- (a) The SRC are selling daffodils for \$5 each. They buy the daffodils for \$3 each and have other costs of \$100. The equation for their sales, S , in terms of x , where x is the total number of daffodils (all daffodils are sold) is $S = 5x$.

- (i) Write an equation for their costs, C , in terms of x .

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- (ii) Find the number of daffodils at the break-even point.

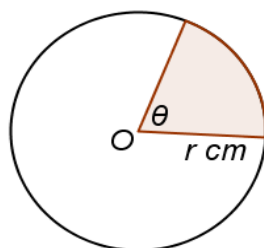
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Question 10 continues on page 13

Question 10 (continued)

- (b) In the circle below with centre O , radius r cm, and angle at the centre of the shaded sector θ radians, the perimeter of the shaded sector is 14 cm and its area is 12 cm^2 .



NOT TO
SCALE

- (i) Show that $\theta = \frac{14 - 2r}{r}$. 1

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- (ii) Show that $\theta = \frac{24}{r^2}$. 1

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- (iii) Find the possible values of the radius of the circle. 3

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End of Question 10

Question 11 (7 marks)

- (a) The area, A , of a pizza slice varies inversely with the number of slices, n , into which it is cut. When it is cut into 8 slices, the area of each slice is 88 cm^2 .

- (i) Find the equation relating A and n .

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- (ii) What is the area of one pizza slice when it is cut into 10 slices?

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- (iii) How many slices are there if each slice has an area of 100.57 cm^2 .

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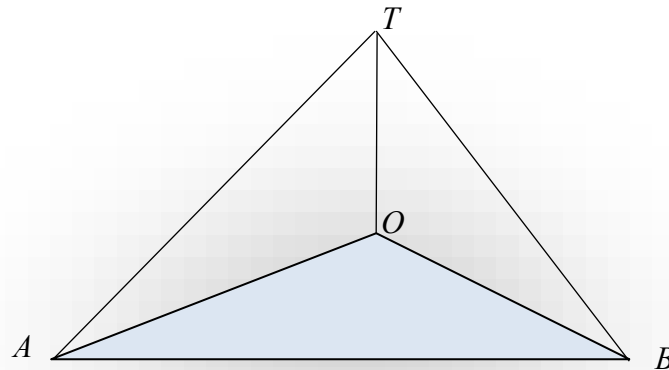
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Question 11 continues on page 15

Question 11 (continued)

- (b) Alex walks along a straight road on level ground. At one point, A , a tower is noticed on a bearing of 053° , with an angle of elevation of 21° . After Alex walks a further 230 m to a point B , the tower is on a bearing of 342° with an angle of elevation of 26° . Let the base of the tower be the point O and the top of the tower be the point T .



- (i) State the size of $\angle AOB$. 1

.....

- (ii) Find the height of the tower correct to the nearest metre. 3

.....

END OF PAPER

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[illegible]

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Mathematics Advanced

Year 11 Assessment 2
Term 2 2023

STUDENT NUMBER: _____

Section I – Multiple Choice Answer Sheet

Allow about 15 minutes for this section

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A ☒ B ☒ ^{correct} C ☐ D ☐

1. A ☐ B ☐ C ☐ D ☐

2. A ☐ B ☐ C ☐ D ☐

3. A ☐ B ☐ C ☐ D ☐

4. A ☐ B ☐ C ☐ D ☐

5. A ☐ B ☐ C ☐ D ☐



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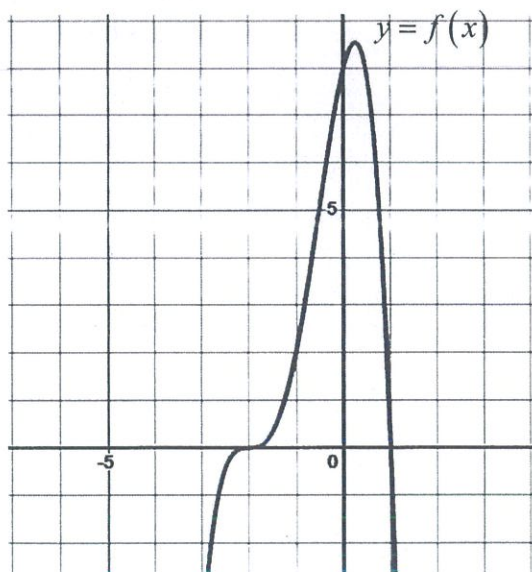
A. $\cos 60^\circ = \sin 30^\circ$

B. $\sin(-30^\circ) = -\sin 30^\circ$

C. $-\sin 210^\circ = -(-\sin 30^\circ)$ 3rd quad.

D. $\sin \frac{5\pi}{6} = \sin \frac{\pi}{6} = \sin 30^\circ$

2 The graph of $y = f(x)$ is shown on the number plane below.



Which of the following could be the equation of this graph?

A. $y = (x-1)(2-x)^3$

B. $y = (x+1)(x+2)^3$ ✗

C. $y = (1-x)(x+2)^3$

D. $y = (1+x)(x-2)^3$ ✗

concave down $\therefore -x^4$
 $\therefore$ not B or D

behaves like a cubic at $x = -2$
 $\therefore (x+2)^3 \therefore C$

3 What are the coordinates of the vertex of the parabola $y = -(x-3)^2 + 2$?

- A. (3, 2)
- B. (3, -2)
- C. (-3, 2)
- D. (-3, -2)

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5 Let $h(x) = f(g(x))$ where the function $f(x)$ is an even function and the function $g(x)$ is an odd function.

Which of the following statements is **true**?

- A. $h(x)$ is an even function.
- B. $h(x)$ is an odd function.
- C. $h(x)$ is both an even and an odd function.
- D. $h(x)$ is neither an even nor an odd function.

$$\begin{aligned}
 h(u) &= f(g(u)) \\
 h(-u) &= f(g(-u)) \\
 &= f(-g(u)) \quad \text{since } g(u) \text{ is odd} \\
 h(-u) &= f(g(u)) \quad \text{since } f(u) \text{ is even} \\
 h(-u) &= h(u) \therefore h(u) \text{ is even}
 \end{aligned}$$

End of Section I

Q 1 B
2 C
3 A
4 D
5 A

Section II

50 marks

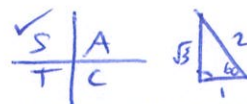
Attempt Questions 6-11

Allow about 52 minutes for this section

Question 6 (8 marks)

- (a) Find the exact value of $\sin 120^\circ$.

$$\begin{aligned} &= \sin(180^\circ - 60^\circ) \\ &= \sin 60^\circ \\ &= \frac{\sqrt{3}}{2} \end{aligned}$$

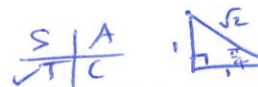


1

Well done!

- (b) Find the exact value of $\cos \frac{5\pi}{4}$.

$$\begin{aligned} &= \cos\left(\pi + \frac{\pi}{4}\right) \\ &= -\cos \frac{\pi}{4} \\ &= -\frac{1}{\sqrt{2}} \end{aligned}$$

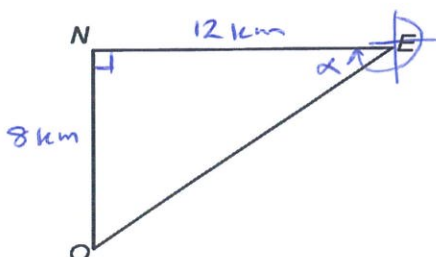


1

Well done!

- (c) A ship sails from a point O due north for 8 km to a point N , and then sails due east for 12 km to a point E . What is the bearing of O from E , to the nearest degree?

2



NOT TO
SCALE

$$\tan \alpha = \frac{8}{12} = \frac{2}{3}$$

$$\alpha = 33.69006753$$

$$\alpha \approx 34^\circ$$

1 for angle

Bearing of O from E is $270 - 34^\circ = 236^\circ$

$$\text{OR } 90 - 34^\circ = 56^\circ \therefore S 56^\circ W$$

1 for bearing
in either form

Question 6 continues on page 5

Generally well done!
Some students forgot
to calculate the
bearing.

(d) Simplify $\tan(2\pi - \theta)$

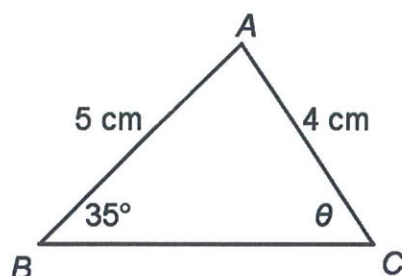
$$\frac{s}{T} \left| \begin{array}{c} A \\ c \end{array} \right.$$

1

$$= -\tan \theta$$

Well done!

(e)



NOT TO
SCALE

Find the size of $\angle ACB$ correct to the nearest degree.

3

$$\frac{\sin \theta}{5} = \frac{\sin 35^\circ}{4}$$

$$\sin \theta = \frac{5 \sin 35^\circ}{4}$$

$$\sin \theta = 0.7169705454$$

$$\theta = 45.80492614^\circ \text{ or } 180^\circ - 45.8^\circ$$

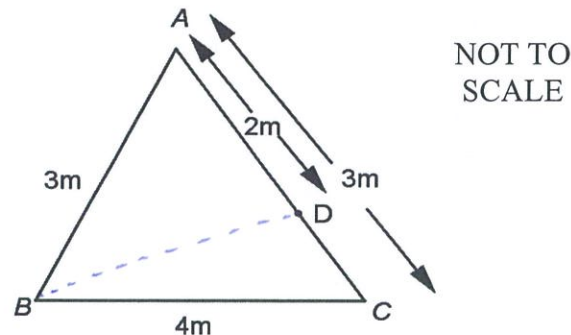
$$\theta = 46^\circ \text{ or } 134^\circ$$

Many students forgot
to check for ambiguous
case.

End of Question 6

Question 7 (9 marks)

- (a) In the diagram $AB = 3$ m, $AC = 3$ m and $BC = 4$ m.
The point D on AC is such that $AD = 2$ m.



- (i) Find the value of $\angle A$, to the nearest minute.

2

$$\cos A = \frac{3^2 + 3^2 - 4^2}{2 \times 3 \times 3}$$

$$= \frac{18 - 16}{18}$$

$$= \frac{2}{18}$$

$$\cos A = \frac{1}{9}$$

$$A = 83.62062979 \div 83^\circ 37'$$

1. Nearest minute is required.

1 for correct application of cosine Rule

* Generally done well.

- (ii) Calculate the length of BD , correct to 2 decimal places.

2

$$BD^2 = 3^2 + 2^2 - 2 \times 3 \times 2 \times \cos A$$

$$= 9 + 4 - 12 \times \frac{1}{9}$$

$$= 13 - \frac{12}{9}$$

$$BD^2 = 11\frac{2}{3}$$

$$BD = \pm \sqrt{11\frac{2}{3}} \text{ but } BD > 0$$

$$BD \div 3.415650255$$

$$BD \div 3.42 \text{ m to 2 decimal places}$$

* Generally done well.

- (iii) Find the area of $\triangle ABD$, correct to 2 decimal places.

2

$$\text{Area} = \frac{1}{2} \times 3 \times 2 \times \sin A$$

$$= 3 \sin 83^\circ 37' \text{ or } 3 \times \frac{\sqrt{80}}{9}$$

$$= 2.981400900$$

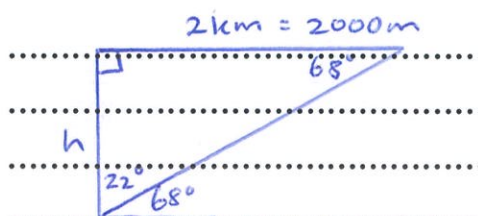
$$= 2.98 \text{ m}^2 \text{ to 2 decimal places}$$

* Many found area of $\triangle ABC$!

Question 7 continues on page 7

- (b) A plane is travelling at a constant speed of 600 km/h. At a certain instant it is directly overhead. Twelve seconds later, its angle of elevation is 68° . Find the height of the plane above the ground to the nearest metre.

3



600 km/h
 $= 600 \text{ km} / 60 \text{ min}$
 $= 10 \text{ km} / \text{min}$
 $= \frac{10}{60} \text{ km} / \text{s}$
 $= 1 \text{ km} / 6 \text{ s}$
 $= 2 \text{ km} / 12 \text{ s}$ (for correct distance travelled in 12 seconds)

* Many did not place angle of elevation in correct position. This was a recurring error.

$\tan 68^\circ = \frac{h}{2000}$ (for correct equation)

$h = 2000 \tan 68^\circ$
 $h = 4950.173707$
 $h = 4950 \text{ m}$ or 4.950 km (for correct evaluation)

* Many concluded that the distance was 5m.

or $\tan 68 = \frac{h}{2}$ or $\tan 22 = \frac{2000}{h}$ or $\tan 22 = \frac{2}{h}$

End of Question 7

Question 8 (9 marks)

- (a) Given that $f(x) = x^2$ and $g(x) = \sqrt{x}$, state the domain of $f(g(x))$.

1

D: all real x

D: x ≥ 0

D: x > 0

Mostly did not know how to find the domain.

- (b) Find the equation of the cubic function $f(x) = kx^3 + c$ if it passes through the point $(-1, 2)$ and has y intercept -3 .

2

y intercept is -3 ∴ passes through (0, -3)

$$f(0) = k \times 0 + c = -3$$

$$c = -3$$

$$\therefore f(x) = kx^3 - 3$$

Passes through (-1, 2)

$$\therefore f(-1) = k(-1)^3 - 3 = 2$$

$$-k - 3 = 2$$

$$k = -5$$

$$\therefore f(x) = -5x^3 - 3$$

Mostly did well.

It's a worry when students used $m = \frac{y_2 - y_1}{x_2 - x_1}$ to find k when it is not a linear function.

- (c) Find the points of intersection of the graphs of $y = x^2 - x$ and $y = 2x - 2$.

3

$$x^2 - x = 2x - 2$$

$$x^2 - 3x + 2 = 0$$

$$(x - 2)(x - 1) = 0$$

$$x = 2 \quad \text{or} \quad x = 1$$

$$y = 2 \quad \quad y = 0$$

1 for method (could do it graphically)

1 for correct x values

The points of intersection are (2, 2) and (1, 0)

1 for both correct solutions

Mostly did well, however it is a worry when some students did not even know how to do this question.

- (d) Find the points of intersection of the graphs of $4x - 3y = 22$ and $3x + 4y = 29$.

$$\begin{array}{l}
 4x - 3y = 22 \quad \text{--- (1)} \\
 3x + 4y = 29 \quad \text{--- (2)} \\
 \hline
 \textcircled{1} \times 3 = \textcircled{3} \quad 12x - 9y = 66 \quad \text{--- (3)} \\
 \textcircled{2} \times 4 = \textcircled{4} \quad 12x + 16y = 117 \quad \text{--- (4)} \\
 \hline
 \textcircled{4} - \textcircled{3} \quad 25y = 50 \\
 y = 2 \\
 \hline
 \text{Sub } y = 2 \text{ into (2)} \\
 3x + 4 \times 2 = 29 \\
 3x + 8 = 29 \\
 3x = 21 \\
 x = 7
 \end{array}$$

1 for method

1.

Point of intersection is $(7, 2)$

1.

Morky did well.

Again it is a worry when some
students did not know how to do
this simultaneous equations at all.

End of Question 8

Question 9 (8 marks)

(a) Solve $\left| \frac{x}{3} - 2 \right| - 1 = 0$.

Generally well done.
Some student only found one answer.

2

$$\begin{aligned} \left| \frac{x}{3} - 2 \right| &= 1 & \text{or} & \left(\frac{x}{3} - 2 \right)^2 = 1^2 \\ \frac{x}{3} - 2 &= \pm 1 & & \frac{x^2}{9} - \frac{4x}{3} + 4 = 1 \\ \frac{x}{3} &= 2 \pm 1 & & x^2 - 12x + 36 = 9 \\ \frac{x}{3} &= 3 \text{ or } 1 & & x^2 - 12x + 27 = 0 \\ x &= 9 \text{ or } 3 & & (x-9)(x-3) = 0 \\ & & & x = 9 \text{ or } 3 \end{aligned}$$

(b) $A(-1, 3)$ is a vertex of the rectangle $ABCD$ whose side AD has equation $x + 2y - 5 = 0$. Find the equation of the side AB .

2

$$\begin{aligned} x + 2y - 5 &= 0 \\ 2y &= -x + 5 \\ y &= -\frac{1}{2}x + \frac{5}{2} \\ \therefore m_{AD} &= -\frac{1}{2} \text{ but } AD \perp AB \text{ adjacent sides of a rectangle} \\ \therefore m_{AB} &= 2 \end{aligned}$$

Many students were able to find the gradient.
Generally well done.

$$\begin{aligned} \text{Equation of } AB & \\ y - y_1 &= m_{AB}(x - x_1) \\ y - 3 &= 2(x - (-1)) \\ y - 3 &= 2(x + 1) \\ y - 3 &= 2x + 2 \\ y &= 2x + 5 \text{ or } 2x - y + 5 = 0 \end{aligned}$$

Question 9 continues on Page 11

Question 9 (continued)

- (c) (i) If $f(x) = \frac{1}{x-2} + 1$, find $f(3)$.

1

$$f(3) = \frac{1}{3-2} + 1$$

$$= \frac{1}{1} + 1$$

$$f(3) = 2$$

Generally well done.

- (ii) State the domain of the function $f(x) = \frac{1}{x-2} + 1$.

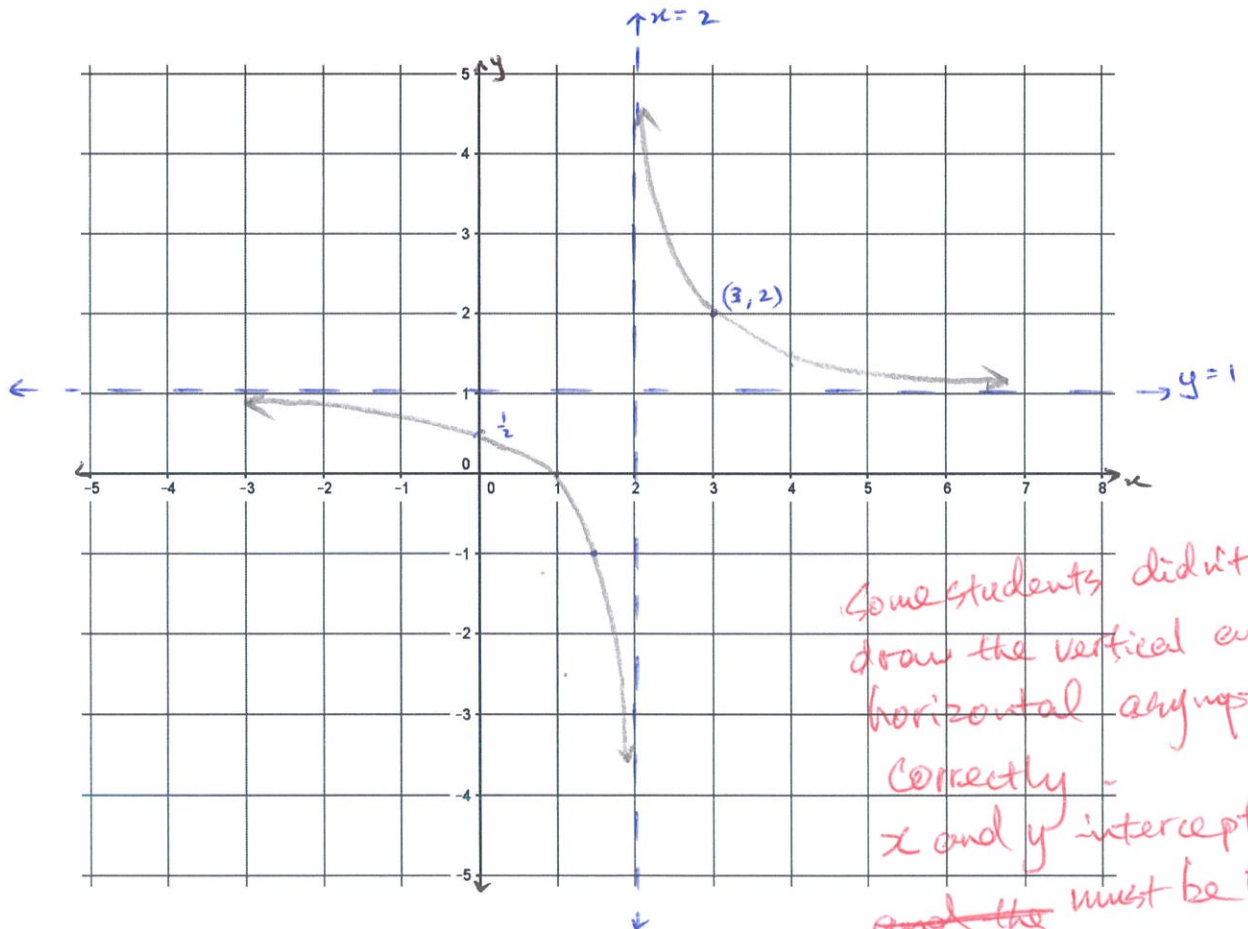
1

$$D: \text{all real } x, x \neq 2 \quad \text{or} \quad (-\infty, 2) \cup (2, \infty)$$

Generally well done

- (iii) Sketch the graph of the function $f(x) = \frac{1}{x-2} + 1$, showing the x and y intercepts and the equations of any asymptotes.

2



Some students didn't draw the vertical and horizontal asymptotes correctly - x and y intercepts ~~and the~~ must be correct. The point $(3, 2)$ is on the curve.

End of Question 9

Question 10 (7 marks)

- (a) The SRC are selling daffodils for \$5 each. They buy the daffodils for \$3 each and have other costs of \$100. The equation for their sales, S , in terms of x , where x is the total number of daffodils (all daffodils are sold) is $S = 5x$.

- (i) Write an equation for their costs, C , in terms of x .

1

..... $C = 3x + 100$ Done well by most students.
.....

- (ii) Find the number of daffodils at the break-even point.

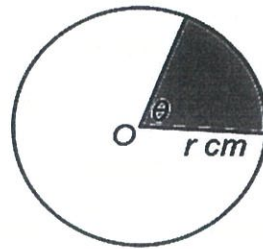
1

..... $5x = 3x + 100$ Some students had no idea
..... $2x = 100$ how to attempt this question
..... $x = 50$
.....
..... There are 50 daffodils at the break-even point.

Question 10 continues on page 13

Question 10 (continued)

- (b) In the circle below with centre O , radius r cm, and angle at the centre of the shaded sector θ radians, the perimeter of the shaded sector is 14 cm and its area is 12 cm^2 .



NOT TO
SCALE

- (i) Show that $\theta = \frac{14-2r}{r}$.

1

$$\begin{aligned}
 P &= 2r + r\theta = 14 & l &= r\theta \\
 r\theta &= 14 - 2r & \theta &= \frac{l}{r} \\
 \theta &= \frac{14-2r}{r} & \theta &= \frac{14-2r}{r} \text{ is insufficient for a 'show that' question}
 \end{aligned}$$

- (ii) Show that $\theta = \frac{24}{r^2}$.

1

$$\begin{aligned}
 A &= \frac{1}{2} r^2 \theta = 12 & \text{Done well by most students} \\
 r^2 \theta &= 24 \\
 \theta &= \frac{24}{r^2}
 \end{aligned}$$

- (iii) Find the possible values of the radius of the circle.

3

$$\begin{aligned}
 \frac{14-2r}{r} &= \frac{24}{r^2} & \text{Some students had no idea} \\
 r^2(14-2r) &= 24 & \text{since } r > 0 \text{ how to approach this question.} \\
 14r - 2r^2 &= 24 & \text{Some tried to substitute the} \\
 0 &= 2r^2 - 14r + 24 & \text{result from part (i) or (ii) back} \\
 0 &= 2(r^2 - 7r + 12) & \text{into an equation from the} \\
 0 &= 2(r-3)(r-4) & \text{same part, which leads} \\
 r &= 3 \text{ cm or } 4 \text{ cm} & \text{no-value.}
 \end{aligned}$$

End of Question 10

Question 11 (7 marks)

- (a) The area, A , of a pizza slice varies inversely with the number of slices, n , into which it is cut. When it is cut into 8 slices, the area of each slice is 88 cm^2 .

- (i) Find the equation relating A and n .

1

$$\begin{aligned}
 A &\propto \frac{1}{n} \\
 A &= \frac{k}{n} \\
 88 &= \frac{k}{8} \\
 k &= 8 \times 88 \\
 k &= 704 \\
 \therefore A &= \frac{704}{n}
 \end{aligned}$$

Mostly well done
however, some students
did not read/ignored 'inversely'

- (ii) What is the area of one pizza slice when it is cut into 10 slices?

1

$$\begin{aligned}
 \text{When } n=10, \quad A &= \frac{704}{10} \\
 A &= 70.4 \text{ cm}^2
 \end{aligned}$$

Mostly well done
follow on errors
were allowed 'c/fpe'

- (iii) How many slices are there if each slice has an area of 100.57 cm^2 .

1

$$\begin{aligned}
 \text{When } A &= 100.57, \quad 100.57 = \frac{704}{n} \\
 n &= \frac{704}{100.57} \\
 n &= 7.000099433 \\
 n &= 7 \text{ slices}
 \end{aligned}$$

Context!
 100.57 is NOT 100×57
A dot is used in
Algebra for times and
it is always written low
not in the middle e.g. $a \times b = a \cdot b$ ✓
 $= a \cdot b \times$

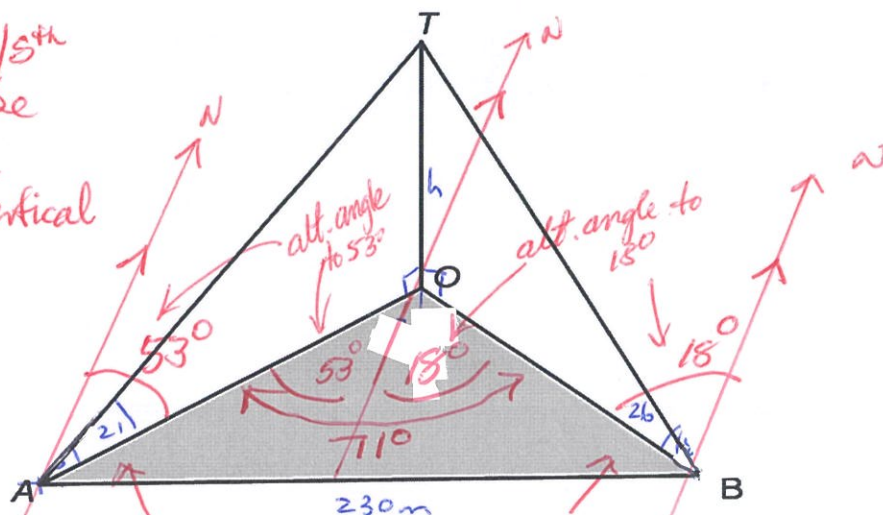
Mostly well
done and
follow on
errors
allowed 'c/fpe'

Question 11 continues on page 15

Question 11 (continued)

- (b) Alex walks along a straight road on level ground. At one point, A , a tower is noticed on a bearing of 053° , with an angle of elevation of 21° . After Alex walks a further 230 m to a point B , the tower is on a bearing of 342° with an angle of elevation of 26° . Let the base of the tower be the point O and the top of the tower be the point T .

In 3D, Nth/Sth lines should be drawn oblique rather than vertical



NOT TO SCALE

- (i) State the size of $\angle AOB$. *These angles could not be written directly from info as AB is NOT East/West*

$$\angle AOB = 53^\circ + 18^\circ = 71^\circ$$

Poorly done
Many assumed AB is an East-West line which it isn't.

- (ii) Find the height of the tower correct to the nearest metre.

3

$$\text{In } \triangle AOT, \tan 21^\circ = \frac{h}{OA}$$

$$OA = \frac{h}{\tan 21^\circ} = h \cot 21^\circ = h \tan 69^\circ$$

$$\text{In } \triangle BOT, \tan 26^\circ = \frac{h}{OB}$$

$$OB = \frac{h}{\tan 26^\circ} = h \cot 26^\circ = h \tan 64^\circ$$

In $\triangle AOB$

$$AB^2 = OA^2 + OB^2 - 2 \times OA \times OB \times \cos(\angle AOB)$$

$$230^2 = h^2 \tan^2 69^\circ + h^2 \tan^2 64^\circ - 2 \times h \tan 69^\circ \times h \tan 64^\circ \times \cos 71^\circ$$

$$230^2 = h^2 (\tan^2 69^\circ + \tan^2 64^\circ - 2 \tan 69^\circ \tan 64^\circ \cos 71^\circ)$$

$$h^2 = \frac{230^2}{\tan^2 69^\circ + \tan^2 64^\circ - 2 \tan 69^\circ \tan 64^\circ \cos 71^\circ}$$

$$h^2 = 7041.719265 \quad \text{but } h > 0$$

$$h = 83.91495257$$

$$h \div 84 \text{ m to the nearest metre.}$$

①

END OF PAPER